

Bragg Diffraction — Raw Experimental Data

Practical Physics II

Notes

All tables below contain unprocessed raw count data recorded using a fixed gate time. Angles are the crystal angle θ_c . These are the counts as recorded by the X-ray unit. I haven't touched them.

Table 1: NaCl (100), Measurement 1 (3.0°–8.0°)

$\theta_c / ^\circ$	Counts
3.0	573
3.1	571
3.2	520
3.3	422
3.4	367
3.5	312
3.6	288
3.7	262
3.8	235
3.9	209
4.0	176
4.1	166
4.2	146
4.3	133
4.4	134
4.5	116
4.6	116
4.7	117
4.8	107
4.9	106
5.0	98
5.1	108
5.2	94
5.3	115
5.4	103
5.5	103
5.6	97
5.7	92
5.8	92
5.9	94
6.0	97
6.1	106
6.2	103
6.3	119
6.4	164
6.5	215
6.6	138
6.7	123
6.8	91
6.9	106
7.0	147
7.1	141
7.2	409
7.3	631
7.4	301
7.5	117
7.6	122
7.7	113
7.8	125
7.9	111
8.0	121

Table 2: NaCl (100), Measurement 2 (3.0°–8.0°)

$\theta_c / ^\circ$	Counts
3.0	210
3.1	217
3.2	244
3.3	227
3.4	195
3.5	169
3.6	147
3.7	131
3.8	119
3.9	136
4.0	120
4.1	108
4.2	110
4.3	107
4.4	112
4.5	90
4.6	121
4.7	84
4.8	100
4.9	109
5.0	83
5.1	92
5.2	107
5.3	92
5.4	96
5.5	96
5.6	98
5.7	92
5.8	99
5.9	87
6.0	100
6.1	142
6.2	148
6.3	159
6.4	186
6.5	173
6.6	114
6.7	109
6.8	105
6.9	187
7.0	474
7.1	386
7.2	463
7.3	454
7.4	123
7.5	131
7.6	125
7.7	125
7.8	120
7.9	135
8.0	131

Table 3: NaCl (100), Measurement 1 (8.1°–13.0°)

$\theta_c / ^\circ$	Counts
8.1	129
8.2	125
8.3	115
8.4	119
8.5	122
8.6	117
8.7	123
8.8	115
8.9	118
9.0	129
9.1	109
9.2	124
9.3	111
9.4	111
9.5	102
9.6	103
9.7	93
9.8	112
9.9	110
10.0	100
10.1	111
10.2	99
10.3	96
10.4	100
10.5	100
10.6	100
10.7	105
10.8	97
10.9	101
11.0	99
11.1	96
11.2	98
11.3	102
11.4	98
11.5	91
11.6	92
11.7	98
11.8	88
11.9	100
12.0	98
12.1	87
12.2	96
12.3	90
12.4	86
12.5	97
12.6	96
12.7	98
12.8	106
12.9	112
13.0	131

Table 4: NaCl (100), Measurement 2 (8.1°–13.0°)

$\theta_c / ^\circ$	Counts
8.1	116
8.2	126
8.3	132
8.4	128
8.5	131
8.6	133
8.7	124
8.8	137
8.9	139
9.0	120
9.1	137
9.2	123
9.3	116
9.4	118
9.5	115
9.6	113
9.7	108
9.8	111
9.9	104
10.0	119
10.1	99
10.2	113
10.3	108
10.4	101
10.5	105
10.6	110
10.7	108
10.8	114
10.9	106
11.0	102
11.1	102
11.2	102
11.3	103
11.4	91
11.5	105
11.6	115
11.7	90
11.8	91
11.9	88
12.0	101
12.1	98
12.2	102
12.3	98
12.4	100
12.5	94
12.6	111
12.7	137
12.8	130
12.9	133
13.0	116

Table 5: NaCl (100), Measurement 1 (13.1°–18.0°)

$\theta_c / ^\circ$	Counts
13.1	126
13.2	97
13.3	95
13.4	83
13.5	93
13.6	91
13.7	88
13.8	88
13.9	101
14.0	97
14.1	87
14.2	81
14.3	94
14.4	104
14.5	153
14.6	289
14.7	289
14.8	126
14.9	89
15.0	81
15.1	78
15.2	84
15.3	88
15.4	76
15.5	78
15.6	65
15.7	71
15.8	67
15.9	75
16.0	72
16.1	52
16.2	65
16.3	65
16.4	63
16.5	60
16.6	55
16.7	70
16.8	53
16.9	50
17.0	51
17.1	50
17.2	62
17.3	67
17.4	53
17.5	51
17.6	41
17.7	55
17.8	61
17.9	55
18.0	54

Table 6: NaCl (100), Measurement 2 (13.1°–18.0°)

$\theta_c / ^\circ$	Counts
13.1	111
13.2	100
13.3	86
13.4	97
13.5	91
13.6	105
13.7	79
13.8	96
13.9	86
14.0	82
14.1	90
14.2	102
14.3	214
14.4	247
14.5	260
14.6	286
14.7	154
14.8	99
14.9	74
15.0	83
15.1	98
15.2	80
15.3	71
15.4	74
15.5	73
15.6	71
15.7	67
15.8	65
15.9	83
16.0	70
16.1	65
16.2	65
16.3	61
16.4	67
16.5	61
16.6	58
16.7	56
16.8	50
16.9	51
17.0	56
17.1	56
17.2	55
17.3	57
17.4	62
17.5	56
17.6	45
17.7	57
17.8	50
17.9	41
18.0	45

Table 7: NaCl (100), Measurement 1 (18.1°–23.0°)

$\theta_c / ^\circ$	Counts
18.1	51
18.2	45
18.3	59
18.4	58
18.5	45
18.6	45
18.7	51
18.8	43
18.9	43
19.0	42
19.1	41
19.2	42
19.3	43
19.4	37
19.5	46
19.6	53
19.7	51
19.8	55
19.9	37
20.0	38
20.1	42
20.2	43
20.3	38
20.4	42
20.5	38
20.6	37
20.7	38
20.8	48
20.9	36
21.0	27
21.1	39
21.2	40
21.3	37
21.4	37
21.5	41
21.6	38
21.7	33
21.8	44
21.9	37
22.0	33
22.1	63
22.2	108
22.3	114
22.4	70
22.5	42
22.6	34
22.7	40
22.8	36
22.9	37
23.0	35

Table 8: NaCl (100), Measurement 2 (18.1°–23.0°)

$\theta_c / ^\circ$	Counts
18.1	39
18.2	39
18.3	43
18.4	55
18.5	47
18.6	47
18.7	54
18.8	45
18.9	47
19.0	42
19.1	47
19.2	45
19.3	46
19.4	49
19.5	54
19.6	58
19.7	58
19.8	38
19.9	29
20.0	34
20.1	44
20.2	39
20.3	35
20.4	41
20.5	33
20.6	34
20.7	30
20.8	33
20.9	27
21.0	29
21.1	34
21.2	40
21.3	37
21.4	31
21.5	33
21.6	32
21.7	35
21.8	43
21.9	75
22.0	94
22.1	120
22.2	103
22.3	79
22.4	37
22.5	27
22.6	30
22.7	34
22.8	31
22.9	25
23.0	34

Table 9: NaCl (100), Measurement 1 (23.1°–28.0°)

$\theta_c / ^\circ$	Counts
23.1	33
23.2	29
23.3	27
23.4	34
23.5	30
23.6	25
23.7	28
23.8	30
23.9	34
24.0	25
24.1	31
24.2	23
24.3	31
24.4	30
24.5	26
24.6	29
24.7	32
24.8	26
24.9	33
25.0	32
25.1	24
25.2	33
25.3	26
25.4	27
25.5	24
25.6	23
25.7	21
25.8	25
25.9	25
26.0	24
26.1	21
26.2	26
26.3	33
26.4	23
26.5	28
26.6	33
26.7	35
26.8	27
26.9	20
27.0	25
27.1	21
27.2	19
27.3	23
27.4	26
27.5	27
27.6	22
27.7	24
27.8	23
27.9	21
28.0	21

Table 10: NaCl (100), Measurement 2 (23.1°–28.0°)

$\theta_c / ^\circ$	Counts
23.1	26
23.2	29
23.3	30
23.4	29
23.5	36
23.6	28
23.7	30
23.8	23
23.9	29
24.0	24
24.1	33
24.2	33
24.3	23
24.4	22
24.5	21
24.6	21
24.7	16
24.8	20
24.9	23
25.0	26
25.1	23
25.2	21
25.3	27
25.4	25
25.5	22
25.6	22
25.7	24
25.8	24
25.9	30
26.0	29
26.1	21
26.2	25
26.3	17
26.4	28
26.5	21
26.6	21
26.7	22
26.8	23
26.9	22
27.0	21
27.1	20
27.2	25
27.3	22
27.4	17
27.5	22
27.6	17
27.7	22
27.8	19
27.9	22
28.0	21

Table 11: NaCl (100), Measurement 1 (28.1°–30.0°)

$\theta_c / ^\circ$	Counts
28.1	22
28.2	22
28.3	26
28.4	13
28.5	22
28.6	17
28.7	23
28.8	17
28.9	22
29.0	24
29.1	18
29.2	17
29.3	24
29.4	22
29.5	12
29.6	25
29.7	16
29.8	22
29.9	19
30.0	23

Table 12: NaCl (100), Measurement 2 (28.1°–30.0°)

$\theta_c / ^\circ$	Counts
28.1	19
28.2	20
28.3	18
28.4	15
28.5	20
28.6	22
28.7	17
28.8	16
28.9	13
29.0	21
29.1	16
29.2	22
29.3	19
29.4	18
29.5	13
29.6	18
29.7	10
29.8	19
29.9	18
30.0	30

Table 13: NaCl (110), Measurement 1 (3.0°–8.0°)

$\theta_c / ^\circ$	Counts
3.0	444
3.1	379
3.2	370
3.3	299
3.4	279
3.5	253
3.6	213
3.7	200
3.8	171
3.9	160
4.0	136
4.1	131
4.2	111
4.3	86
4.4	86
4.5	107
4.6	84
4.7	95
4.8	111
4.9	103
5.0	97
5.1	108
5.2	91
5.3	119
5.4	96
5.5	106
5.6	110
5.7	115
5.8	117
5.9	104
6.0	96
6.1	115
6.2	111
6.3	135
6.4	115
6.5	119
6.6	116
6.7	116
6.8	106
6.9	117
7.0	108
7.1	105
7.2	101
7.3	122
7.4	97
7.5	109
7.6	108
7.7	123
7.8	111
7.9	118
8.0	106

Table 14: NaCl (110), Measurement 2 (3.0°–8.0°)

$\theta_c / ^\circ$	Counts
3.0	321
3.1	334
3.2	301
3.3	320
3.4	266
3.5	248
3.6	216
3.7	189
3.8	172
3.9	154
4.0	122
4.1	125
4.2	109
4.3	105
4.4	115
4.5	100
4.6	109
4.7	101
4.8	110
4.9	96
5.0	99
5.1	109
5.2	117
5.3	107
5.4	135
5.5	126
5.6	122
5.7	108
5.8	108
5.9	104
6.0	114
6.1	124
6.2	116
6.3	107
6.4	110
6.5	116
6.6	103
6.7	114
6.8	104
6.9	116
7.0	111
7.1	121
7.2	103
7.3	96
7.4	110
7.5	107
7.6	109
7.7	97
7.8	97
7.9	106
8.0	102

Table 15: NaCl (110), Measurement 1 (8.1°–13.0°)

$\theta_c / ^\circ$	Counts
8.1	100
8.2	104
8.3	98
8.4	105
8.5	102
8.6	104
8.7	103
8.8	87
8.9	109
9.0	128
9.1	214
9.2	210
9.3	131
9.4	97
9.5	91
9.6	107
9.7	99
9.8	111
9.9	102
10.0	142
10.1	189
10.2	498
10.3	743
10.4	473
10.5	172
10.6	113
10.7	97
10.8	96
10.9	101
11.0	100
11.1	103
11.2	95
11.3	105
11.4	100
11.5	107
11.6	91
11.7	92
11.8	89
11.9	94
12.0	87
12.1	75
12.2	87
12.3	79
12.4	102
12.5	85
12.6	76
12.7	73
12.8	81
12.9	87
13.0	82

Table 16: NaCl (110), Measurement 2 (8.1°–13.0°)

$\theta_c / ^\circ$	Counts
8.1	108
8.2	97
8.3	101
8.4	94
8.5	98
8.6	87
8.7	104
8.8	85
8.9	119
9.0	175
9.1	224
9.2	181
9.3	115
9.4	103
9.5	95
9.6	99
9.7	108
9.8	107
9.9	117
10.0	147
10.1	228
10.2	589
10.3	665
10.4	306
10.5	121
10.6	101
10.7	98
10.8	107
10.9	92
11.0	86
11.1	91
11.2	107
11.3	91
11.4	83
11.5	83
11.6	86
11.7	88
11.8	79
11.9	92
12.0	94
12.1	92
12.2	79
12.3	81
12.4	88
12.5	77
12.6	68
12.7	82
12.8	76
12.9	80
13.0	79

Table 17: NaCl (110), Measurement 1 (13.1°–18.0°)

$\theta_c / ^\circ$	Counts
13.1	84
13.2	77
13.3	78
13.4	80
13.5	76
13.6	79
13.7	75
13.8	73
13.9	66
14.0	75
14.1	71
14.2	72
14.3	62
14.4	76
14.5	70
14.6	69
14.7	63
14.8	64
14.9	59
15.0	67
15.1	59
15.2	59
15.3	64
15.4	47
15.5	64
15.6	59
15.7	69
15.8	55
15.9	62
16.0	54
16.1	49
16.2	54
16.3	48
16.4	57
16.5	48
16.6	55
16.7	52
16.8	51
16.9	47
17.0	52
17.1	46
17.2	43
17.3	42
17.4	41
17.5	43
17.6	37
17.7	51
17.8	34
17.9	46
18.0	45

Table 18: NaCl (110), Measurement 2 (13.1°–18.0°)

$\theta_c / ^\circ$	Counts
13.1	72
13.2	71
13.3	78
13.4	69
13.5	64
13.6	57
13.7	65
13.8	73
13.9	73
14.0	56
14.1	71
14.2	63
14.3	65
14.4	59
14.5	59
14.6	65
14.7	61
14.8	50
14.9	62
15.0	59
15.1	54
15.2	50
15.3	60
15.4	47
15.5	62
15.6	49
15.7	60
15.8	63
15.9	59
16.0	60
16.1	45
16.2	46
16.3	45
16.4	51
16.5	48
16.6	50
16.7	45
16.8	53
16.9	49
17.0	52
17.1	48
17.2	35
17.3	50
17.4	34
17.5	38
17.6	37
17.7	39
17.8	37
17.9	33
18.0	40

Table 19: NaCl (110), Measurement 1 (18.1°–23.0°)

$\theta_c / ^\circ$	Counts
18.1	47
18.2	38
18.3	46
18.4	61
18.5	53
18.6	58
18.7	44
18.8	44
18.9	38
19.0	43
19.1	35
19.2	43
19.3	45
19.4	33
19.5	36
19.6	28
19.7	31
19.8	27
19.9	34
20.0	32
20.1	41
20.2	38
20.3	34
20.4	34
20.5	33
20.6	49
20.7	76
20.8	129
20.9	128
21.0	87
21.1	56
21.2	28
21.3	39
21.4	35
21.5	29
21.6	35
21.7	23
21.8	27
21.9	30
22.0	21
22.1	23
22.2	29
22.3	26
22.4	24
22.5	25
22.6	27
22.7	29
22.8	23
22.9	27
23.0	26

Table 20: NaCl (110), Measurement 2 (18.1°–23.0°)

$\theta_c / ^\circ$	Counts
18.1	43
18.2	43
18.3	49
18.4	51
18.5	55
18.6	36
18.7	29
18.8	38
18.9	35
19.0	36
19.1	32
19.2	34
19.3	46
19.4	33
19.5	24
19.6	29
19.7	32
19.8	37
19.9	31
20.0	35
20.1	46
20.2	33
20.3	31
20.4	32
20.5	38
20.6	43
20.7	58
20.8	95
20.9	89
21.0	56
21.1	38
21.2	34
21.3	23
21.4	28
21.5	28
21.6	26
21.7	23
21.8	23
21.9	25
22.0	24
22.1	22
22.2	24
22.3	26
22.4	26
22.5	21
22.6	23
22.7	23
22.8	24
22.9	24
23.0	25

Table 21: NaCl (110), Measurement 1 (23.1°–28.0°)

$\theta_c / ^\circ$	Counts
23.1	25
23.2	24
23.3	23
23.4	22
23.5	26
23.6	17
23.7	19
23.8	20
23.9	24
24.0	16
24.1	21
24.2	22
24.3	19
24.4	23
24.5	18
24.6	20
24.7	22
24.8	20
24.9	17
25.0	21
25.1	14
25.2	20
25.3	17
25.4	19
25.5	17
25.6	17
25.7	25
25.8	18
25.9	12
26.0	18
26.1	18
26.2	23
26.3	16
26.4	20
26.5	23
26.6	29
26.7	30
26.8	21
26.9	21
27.0	20
27.1	14
27.2	14
27.3	14
27.4	16
27.5	18
27.6	15
27.7	16
27.8	16
27.9	16
28.0	15

Table 22: NaCl (110), Measurement 2 (23.1°–28.0°)

$\theta_c / ^\circ$	Counts
23.1	22
23.2	15
23.3	21
23.4	23
23.5	21
23.6	17
23.7	22
23.8	19
23.9	13
24.0	16
24.1	20
24.2	20
24.3	20
24.4	14
24.5	25
24.6	18
24.7	15
24.8	19
24.9	15
25.0	12
25.1	15
25.2	16
25.3	17
25.4	11
25.5	15
25.6	20
25.7	14
25.8	15
25.9	17
26.0	15
26.1	15
26.2	11
26.3	18
26.4	21
26.5	22
26.6	29
26.7	26
26.8	20
26.9	13
27.0	15
27.1	19
27.2	11
27.3	23
27.4	12
27.5	19
27.6	16
27.7	10
27.8	14
27.9	18
28.0	16

Table 23: NaCl (110), Measurement 1 (28.1°–30.0°)

$\theta_c / ^\circ$	Counts
28.1	12
28.2	17
28.3	20
28.4	20
28.5	11
28.6	17
28.7	16
28.8	19
28.9	17
29.0	14
29.1	11
29.2	13
29.3	15
29.4	17
29.5	13
29.6	13
29.7	17
29.8	12
29.9	11
30.0	11

Table 24: NaCl (110), Measurement 2 (28.1°–30.0°)

$\theta_c / ^\circ$	Counts
28.1	13
28.2	15
28.3	14
28.4	12
28.5	15
28.6	14
28.7	14
28.8	12
28.9	15
29.0	8
29.1	13
29.2	11
29.3	11
29.4	14
29.5	11
29.6	11
29.7	11
29.8	13
29.9	11
30.0	12

Table 25: NaCl (111), Measurement 1 (3.0°–8.0°)

$\theta_c / ^\circ$	Counts
3.0	7319
3.1	6614
3.2	4899
3.3	3056
3.4	1660
3.5	1135
3.6	913
3.7	711
3.8	575
3.9	471
4.0	450
4.1	394
4.2	384
4.3	343
4.4	312
4.5	275
4.6	248
4.7	231
4.8	187
4.9	182
5.0	171
5.1	179
5.2	177
5.3	164
5.4	173
5.5	214
5.6	180
5.7	152
5.8	185
5.9	204
6.0	181
6.1	218
6.2	318
6.3	313
6.4	212
6.5	188
6.6	177
6.7	181
6.8	189
6.9	196
7.0	197
7.1	204
7.2	206
7.3	199
7.4	213
7.5	202
7.6	217
7.7	208
7.8	210
7.9	244
8.0	233

Table 26: NaCl (111), Measurement 2 (3.0°–8.0°)

$\theta_c / ^\circ$	Counts
3.0	6206
3.1	5055
3.2	3446
3.3	1723
3.4	1136
3.5	910
3.6	713
3.7	572
3.8	523
3.9	466
4.0	410
4.1	370
4.2	344
4.3	292
4.4	276
4.5	242
4.6	224
4.7	184
4.8	185
4.9	182
5.0	168
5.1	160
5.2	163
5.3	160
5.4	172
5.5	185
5.6	173
5.7	165
5.8	190
5.9	207
6.0	159
6.1	178
6.2	250
6.3	245
6.4	160
6.5	161
6.6	166
6.7	174
6.8	167
6.9	164
7.0	192
7.1	177
7.2	167
7.3	169
7.4	186
7.5	193
7.6	184
7.7	176
7.8	182
7.9	168
8.0	179

Table 27: NaCl (111), Measurement 1 (8.1°–13.0°)

$\theta_c / ^\circ$	Counts
8.1	227
8.2	246
8.3	230
8.4	197
8.5	246
8.6	211
8.7	223
8.8	211
8.9	230
9.0	217
9.1	220
9.2	217
9.3	230
9.4	242
9.5	234
9.6	243
9.7	226
9.8	229
9.9	232
10.0	216
10.1	209
10.2	205
10.3	202
10.4	204
10.5	195
10.6	201
10.7	220
10.8	217
10.9	304
11.0	308
11.1	462
11.2	569
11.3	298
11.4	199
11.5	210
11.6	207
11.7	205
11.8	194
11.9	195
12.0	233
12.1	257
12.2	261
12.3	346
12.4	488
12.5	1294
12.6	1879
12.7	882
12.8	332
12.9	228
13.0	177

Table 28: NaCl (111), Measurement 2 (8.1°–13.0°)

$\theta_c / ^\circ$	Counts
8.1	177
8.2	163
8.3	184
8.4	186
8.5	177
8.6	187
8.7	179
8.8	185
8.9	188
9.0	172
9.1	188
9.2	201
9.3	209
9.4	187
9.5	159
9.6	184
9.7	193
9.8	189
9.9	196
10.0	175
10.1	163
10.2	167
10.3	179
10.4	174
10.5	191
10.6	168
10.7	176
10.8	204
10.9	246
11.0	269
11.1	405
11.2	478
11.3	234
11.4	181
11.5	178
11.6	180
11.7	182
11.8	187
11.9	185
12.0	211
12.1	214
12.2	232
12.3	221
12.4	406
12.5	1173
12.6	1608
12.7	661
12.8	236
12.9	164
13.0	170

Table 29: NaCl (111), Measurement 1 (13.1°–18.0°)

$\theta_c / ^\circ$	Counts
13.1	197
13.2	165
13.3	176
13.4	167
13.5	162
13.6	161
13.7	164
13.8	161
13.9	175
14.0	174
14.1	173
14.2	164
14.3	173
14.4	150
14.5	157
14.6	156
14.7	167
14.8	160
14.9	164
15.0	149
15.1	148
15.2	152
15.3	137
15.4	138
15.5	150
15.6	149
15.7	147
15.8	125
15.9	135
16.0	134
16.1	124
16.2	145
16.3	136
16.4	121
16.5	131
16.6	133
16.7	128
16.8	130
16.9	122
17.0	112
17.1	110
17.2	106
17.3	106
17.4	104
17.5	109
17.6	98
17.7	96
17.8	91
17.9	97
18.0	82

Table 30: NaCl (111), Measurement 2 (13.1°–18.0°)

$\theta_c / ^\circ$	Counts
13.1	164
13.2	153
13.3	152
13.4	159
13.5	160
13.6	143
13.7	145
13.8	140
13.9	143
14.0	140
14.1	133
14.2	147
14.3	153
14.4	135
14.5	128
14.6	155
14.7	136
14.8	158
14.9	133
15.0	137
15.1	127
15.2	131
15.3	124
15.4	133
15.5	123
15.6	120
15.7	118
15.8	127
15.9	120
16.0	128
16.1	131
16.2	125
16.3	116
16.4	110
16.5	104
16.6	107
16.7	102
16.8	101
16.9	118
17.0	102
17.1	110
17.2	101
17.3	93
17.4	85
17.5	89
17.6	94
17.7	101
17.8	86
17.9	90
18.0	83

Table 31: NaCl (111), Measurement 1 (18.1°–23.0°)

$\theta_c / ^\circ$	Counts
18.1	91
18.2	79
18.3	88
18.4	79
18.5	82
18.6	82
18.7	76
18.8	84
18.9	86
19.0	78
19.1	110
19.2	101
19.3	88
19.4	69
19.5	68
19.6	64
19.7	61
19.8	65
19.9	69
20.0	68
20.1	63
20.2	51
20.3	57
20.4	53
20.5	48
20.6	41
20.7	48
20.8	54
20.9	62
21.0	49
21.1	48
21.2	48
21.3	49
21.4	56
21.5	45
21.6	40
21.7	40
21.8	43
21.9	44
22.0	47
22.1	46
22.2	46
22.3	49
22.4	40
22.5	51
22.6	47
22.7	42
22.8	42
22.9	66
23.0	63

Table 32: NaCl (111), Measurement 2 (18.1°–23.0°)

$\theta_c / ^\circ$	Counts
18.1	79
18.2	87
18.3	87
18.4	70
18.5	70
18.6	75
18.7	73
18.8	75
18.9	67
19.0	87
19.1	97
19.2	79
19.3	68
19.4	65
19.5	67
19.6	59
19.7	52
19.8	63
19.9	64
20.0	58
20.1	52
20.2	55
20.3	59
20.4	57
20.5	51
20.6	45
20.7	39
20.8	52
20.9	51
21.0	55
21.1	52
21.2	50
21.3	51
21.4	51
21.5	48
21.6	41
21.7	45
21.8	44
21.9	45
22.0	42
22.1	47
22.2	39
22.3	43
22.4	49
22.5	45
22.6	40
22.7	40
22.8	54
22.9	60
23.0	38

Table 33: NaCl (111), Measurement 1 (23.1°–28.0°)

$\theta_c / ^\circ$	Counts
23.1	44
23.2	47
23.3	35
23.4	36
23.5	37
23.6	35
23.7	37
23.8	30
23.9	37
24.0	32
24.1	29
24.2	33
24.3	33
24.4	33
24.5	31
24.6	33
24.7	31
24.8	34
24.9	30
25.0	35
25.1	24
25.2	28
25.3	27
25.4	29
25.5	27
25.6	25
25.7	25
25.8	36
25.9	65
26.0	55
26.1	44
26.2	32
26.3	28
26.4	20
26.5	23
26.6	22
26.7	23
26.8	19
26.9	25
27.0	26
27.1	27
27.2	18
27.3	20
27.4	20
27.5	15
27.6	21
27.7	23
27.8	15
27.9	23
28.0	20

Table 34: NaCl (111), Measurement 2 (23.1°–28.0°)

$\theta_c / ^\circ$	Counts
23.1	34
23.2	35
23.3	37
23.4	32
23.5	32
23.6	37
23.7	27
23.8	30
23.9	28
24.0	32
24.1	35
24.2	30
24.3	31
24.4	25
24.5	32
24.6	27
24.7	31
24.8	28
24.9	29
25.0	25
25.1	29
25.2	25
25.3	25
25.4	28
25.5	20
25.6	25
25.7	27
25.8	34
25.9	55
26.0	42
26.1	30
26.2	30
26.3	29
26.4	20
26.5	19
26.6	23
26.7	24
26.8	19
26.9	14
27.0	23
27.1	20
27.2	24
27.3	17
27.4	15
27.5	19
27.6	19
27.7	18
27.8	20
27.9	15
28.0	22

Table 35: NaCl (111), Measurement 1 (28.1°–30.0°)

$\theta_c / ^\circ$	Counts
28.1	14
28.2	12
28.3	20
28.4	15
28.5	15
28.6	17
28.7	25
28.8	26
28.9	28
29.0	22
29.1	21
29.2	16
29.3	14
29.4	16
29.5	11
29.6	14
29.7	24
29.8	11
29.9	17
30.0	15

Table 36: NaCl (111), Measurement 2 (28.1°–30.0°)

$\theta_c / ^\circ$	Counts
28.1	15
28.2	16
28.3	18
28.4	17
28.5	15
28.6	17
28.7	13
28.8	19
28.9	19
29.0	24
29.1	18
29.2	18
29.3	16
29.4	12
29.5	15
29.6	12
29.7	11
29.8	13
29.9	15
30.0	21

Table 37: LiF (3.0°–8.0°)

$\theta_c / ^\circ$	Counts
3.0	270
3.1	254
3.2	217
3.3	190
3.4	158
3.5	138
3.6	139
3.7	97
3.8	87
3.9	67
4.0	62
4.1	61
4.2	47
4.3	45
4.4	42
4.5	37
4.6	44
4.7	47
4.8	44
4.9	39
5.0	46
5.1	52
5.2	46
5.3	64
5.4	81
5.5	107
5.6	110
5.7	156
5.8	168
5.9	170
6.0	191
6.1	207
6.2	214
6.3	221
6.4	246
6.5	224
6.6	247
6.7	239
6.8	259
6.9	255
7.0	254
7.1	282
7.2	260
7.3	299
7.4	304
7.5	292
7.6	295
7.7	294
7.8	303
7.9	290
8.0	318

Table 38: KBr (3.0°–8.0°)

$\theta_c / ^\circ$	Counts
3.0	254
3.1	237
3.2	212
3.3	185
3.4	155
3.5	138
3.6	133
3.7	107
3.8	125
3.9	94
4.0	105
4.1	88
4.2	85
4.3	65
4.4	67
4.5	75
4.6	84
4.7	66
4.8	70
4.9	75
5.0	85
5.1	83
5.2	86
5.3	102
5.4	135
5.5	138
5.6	189
5.7	98
5.8	106
5.9	104
6.0	158
6.1	298
6.2	466
6.3	491
6.4	128
6.5	107
6.6	106
6.7	94
6.8	113
6.9	101
7.0	105
7.1	118
7.2	110
7.3	114
7.4	112
7.5	118
7.6	111
7.7	122
7.8	138
7.9	132
8.0	152

Table 39: LiF (8.1°–13.0°)

$\theta_c / ^\circ$	Counts
8.1	329
8.2	319
8.3	314
8.4	350
8.5	356
8.6	331
8.7	348
8.8	384
8.9	514
9.0	582
9.1	813
9.2	969
9.3	742
9.4	487
9.5	383
9.6	379
9.7	386
9.8	390
9.9	538
10.0	948
10.1	1278
10.2	2084
10.3	3262
10.4	2829
10.5	1317
10.6	631
10.7	414
10.8	421
10.9	451
11.0	433
11.1	418
11.2	443
11.3	421
11.4	437
11.5	438
11.6	476
11.7	457
11.8	445
11.9	428
12.0	460
12.1	451
12.2	458
12.3	474
12.4	421
12.5	426
12.6	436
12.7	428
12.8	438
12.9	443
13.0	443

Table 40: KBr (8.1°–13.0°)

$\theta_c / ^\circ$	Counts
8.1	173
8.2	229
8.3	218
8.4	220
8.5	209
8.6	232
8.7	207
8.8	220
8.9	216
9.0	218
9.1	216
9.2	205
9.3	223
9.4	230
9.5	213
9.6	207
9.7	225
9.8	215
9.9	219
10.0	195
10.1	211
10.2	205
10.3	191
10.4	213
10.5	202
10.6	198
10.7	179
10.8	185
10.9	206
11.0	201
11.1	234
11.2	227
11.3	226
11.4	211
11.5	211
11.6	181
11.7	182
11.8	173
11.9	182
12.0	207
12.1	191
12.2	190
12.3	219
12.4	329
12.5	371
12.6	241
12.7	175
12.8	172
12.9	183
13.0	203

Table 41: LiF (13.1°–18.0°)

$\theta_c / ^\circ$	Counts
13.1	442
13.2	417
13.3	411
13.4	403
13.5	366
13.6	353
13.7	360
13.8	373
13.9	362
14.0	371
14.1	372
14.2	353
14.3	370
14.4	361
14.5	368
14.6	343
14.7	352
14.8	361
14.9	359
15.0	323
15.1	341
15.2	334
15.3	321
15.4	310
15.5	332
15.6	333
15.7	330
15.8	335
15.9	284
16.0	310
16.1	348
16.2	304
16.3	296
16.4	299
16.5	296
16.6	310
16.7	290
16.8	263
16.9	272
17.0	296
17.1	270
17.2	259
17.3	277
17.4	263
17.5	251
17.6	275
17.7	266
17.8	216
17.9	273
18.0	256

Table 42: KBr (13.1°–18.0°)

$\theta_c / ^\circ$	Counts
13.1	168
13.2	154
13.3	158
13.4	151
13.5	153
13.6	129
13.7	120
13.8	136
13.9	137
14.0	131
14.1	130
14.2	113
14.3	105
14.4	118
14.5	119
14.6	128
14.7	106
14.8	109
14.9	93
15.0	98
15.1	85
15.2	94
15.3	95
15.4	93
15.5	100
15.6	90
15.7	97
15.8	98
15.9	88
16.0	87
16.1	80
16.2	90
16.3	102
16.4	109
16.5	95
16.6	98
16.7	111
16.8	108
16.9	113
17.0	97
17.1	102
17.2	95
17.3	107
17.4	97
17.5	92
17.6	94
17.7	84
17.8	110
17.9	99
18.0	99

Table 43: LiF (18.1°–23.0°)

$\theta_c / ^\circ$	Counts
18.1	278
18.2	290
18.3	299
18.4	344
18.5	342
18.6	298
18.7	271
18.8	284
18.9	286
19.0	259
19.1	267
19.2	246
19.3	248
19.4	252
19.5	219
19.6	239
19.7	232
19.8	215
19.9	235
20.0	236
20.1	218
20.2	219
20.3	219
20.4	239
20.5	293
20.6	364
20.7	536
20.8	676
20.9	590
21.0	401
21.1	258
21.2	214
21.3	202
21.4	195
21.5	188
21.6	210
21.7	235
21.8	210
21.9	189
22.0	140
22.1	185
22.2	165
22.3	143
22.4	142
22.5	148
22.6	156
22.7	151
22.8	141
22.9	155
23.0	130

Table 44: KBr (18.1°–23.0°)

$\theta_c / ^\circ$	Counts
18.1	105
18.2	103
18.3	90
18.4	97
18.5	96
18.6	93
18.7	109
18.8	130
18.9	133
19.0	118
19.1	88
19.2	85
19.3	81
19.4	93
19.5	82
19.6	99
19.7	86
19.8	89
19.9	101
20.0	89
20.1	75
20.2	94
20.3	94
20.4	94
20.5	93
20.6	87
20.7	84
20.8	92
20.9	80
21.0	93
21.1	81
21.2	83
21.3	86
21.4	79
21.5	85
21.6	75
21.7	76
21.8	87
21.9	80
22.0	78
22.1	83
22.2	90
22.3	85
22.4	83
22.5	86
22.6	99
22.7	78
22.8	87
22.9	81
23.0	82

Table 45: LiF (23.1°–28.0°)

$\theta_c / ^\circ$	Counts
23.1	127
23.2	134
23.3	128
23.4	120
23.5	114
23.6	120
23.7	120
23.8	106
23.9	119
24.0	93
24.1	107
24.2	121
24.3	115
24.4	112
24.5	106
24.6	105
24.7	102
24.8	94
24.9	98
25.0	86
25.1	99
25.2	84
25.3	90
25.4	81
25.5	91
25.6	79
25.7	75
25.8	77
25.9	70
26.0	80
26.1	70
26.2	79
26.3	76
26.4	63
26.5	63
26.6	71
26.7	60
26.8	64
26.9	61
27.0	55
27.1	65
27.2	69
27.3	61
27.4	63
27.5	67
27.6	55
27.7	57
27.8	55
27.9	42
28.0	47

Table 46: KBr (23.1°–28.0°)

$\theta_c / ^\circ$	Counts
23.1	91
23.2	84
23.3	81
23.4	85
23.5	86
23.6	84
23.7	89
23.8	84
23.9	76
24.0	78
24.1	75
24.2	83
24.3	79
24.4	65
24.5	74
24.6	78
24.7	75
24.8	84
24.9	79
25.0	80
25.1	92
25.2	89
25.3	91
25.4	71
25.5	99
25.6	84
25.7	90
25.8	83
25.9	71
26.0	81
26.1	74
26.2	82
26.3	75
26.4	76
26.5	84
26.6	81
26.7	77
26.8	81
26.9	74
27.0	81
27.1	86
27.2	70
27.3	65
27.4	76
27.5	78
27.6	80
27.7	80
27.8	83
27.9	74
28.0	78

Table 47: LiF (28.1°–30.0°)

$\theta_c / ^\circ$	Counts
28.1	59
28.2	65
28.3	63
28.4	47
28.5	46
28.6	48
28.7	51
28.8	44
28.9	46
29.0	52
29.1	45
29.2	42
29.3	35
29.4	45
29.5	39
29.6	39
29.7	44
29.8	40
29.9	37
30.0	40

Table 48: KBr (28.1°–30.0°)

$\theta_c / ^\circ$	Counts
28.1	79
28.2	70
28.3	75
28.4	80
28.5	66
28.6	80
28.7	76
28.8	84
28.9	72
29.0	86
29.1	79
29.2	71
29.3	86
29.4	62
29.5	72
29.6	76
29.7	71
29.8	72
29.9	83
30.0	73